DBMS TASK

ADVANCED SQL QUERY IMPLEMENTATION

OBJECTIVE

1. To implement subqueries and nested queries.
2. To find products whose prices are above the average product price.
3. To identify customers with maximum purchases.
4. To generate complex business queries using SQL.
5. To prepare an advanced SQL report using aggregate functions and joins.

QUERY 1: Find Products Above Average Price

Objective: To identify products whose prices are greater than the average price of all products.

SQL Query:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE > (
  4      SELECT AVG(PRICE)
  5      FROM PRODUCT
  6  );

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK  SELLER_ID
---------- ---------- ---------- ----------
       101
Laptop
         1      55000         10        201

       102
Smartphone
         1      25000         20        201

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
```

QUERY 2: Identify Customers with Maximum Purchases

Objective: To find the customer who has spent the highest total amount on purchases.

SQL Query:

```
SQL> SELECT CUSTOMER_ID, SUM(TOTAL_AMOUNT) AS TOTAL_PURCHASE
  2  FROM ORDERS
  3  GROUP BY CUSTOMER_ID
  4  HAVING SUM(TOTAL_AMOUNT) = (
  5      SELECT MAX(TOTAL)
  6      FROM (
  7          SELECT SUM(TOTAL_AMOUNT) AS TOTAL
  8          FROM ORDERS
  9          GROUP BY CUSTOMER_ID
 10      )
 11  );

CUSTOMER_ID TOTAL_PURCHASE
----------- --------------
        101           4300
```

QUERY 3: Find Best-Selling Products

Objective: To identify the best-selling products based on the total quantity sold.

SQL Query:

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(O.QUANTITY) AS TOTAL_SOLD
  4  FROM PRODUCT P
  5  JOIN ORDER_ITEM O
  6  ON P.PRODUCT_ID = O.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_SOLD
----------
       102
Smartphone
         1

       105
T-Shirt
         1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_SOLD
----------

       104
Smart Watch
         1

       103
Headphones

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_SOLD
----------
         1
```

QUERY 4: Find Customers with Successful Payments

Objective: To display customers who have successfully completed their payments.

SQL Query:

```
SQL> SELECT DISTINCT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  WHERE EXISTS (
  8      SELECT 1
  9      FROM PAYMENT P
 10      WHERE P.ORDER_ID = O.ORDER_ID
 11      AND P.PAYMENT_STATUS = 'Successful'
 12  );

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
        101 Arun
Kumar

        102 Priya
Sharma

        105 Karthik
Raj


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
        107 Vignesh
Mohan
```

QUERY 5: Advanced Customer Purchase Report

Objective: To generate a customer purchase report showing total orders, total purchase amount and average order value.

SQL Query

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         COUNT(O.ORDER_ID) AS TOTAL_ORDERS,
  5         SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE,
  6         AVG(O.TOTAL_AMOUNT) AS AVG_ORDER_VALUE
  7  FROM CUSTOMER C
  8  JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  GROUP BY C.CUSTOMER_ID,
 11           C.FIRST_NAME,
 12           C.LAST_NAME
 13  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_PURCHASE
-------------------------------------------------- ------------ --------------
AVG_ORDER_VALUE
---------------
        101 Arun
Kumar                                                         2           4300
           2150

        106 Divya
Nair                                                          1           3500
           3500

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_PURCHASE
-------------------------------------------------- ------------ --------------
AVG_ORDER_VALUE
---------------

        104 Sneha
Reddy                                                         1           3000
           3000

        102 Priya
Sharma                                                        1           2500

CUSTOMER_ID FIRST_NAME
```

QUERY 6: Find Products That Have Never Been Ordered

Objective: To identify products that have never been included in any customer order.

SQL Query:

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME
  2  FROM PRODUCT P
  3  WHERE NOT EXISTS (
  4      SELECT 1
  5      FROM ORDER_ITEM O
  6      WHERE O.PRODUCT_ID = P.PRODUCT_ID
  7  );

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
       107
Python Programming Book

       108
Data Science Book

       109
Mixer Grinder


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
       110
Electric Kettle

       101
Laptop

       106
Jeans


6 rows selected.
```

CONCLUSION

Thus, the Advanced SQL Query Implementation was successfully performed using the Amazon E-Commerce Database. Subqueries, nested queries, JOIN, EXISTS, NOT EXISTS and aggregate functions were implemented to analyze product sales, customer purchases and payment details and generate meaningful business reports.